

System Detective

Computer Organisation & Operating Systems — System Dashboard

Refresh Data

Run Locality Experiment

CPU / ISA

Architecture: x86_64

Model: 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz

Logical CPUs: 8

Physical Cores: 4

Threads/Core: 2

Memory Usage

Total: 3.7Gi

Used: 606Mi

Free: 2.9Gi

Available: 3.1Gi

Top 5 Processes

PID	Process	State	CPU %	Memory %
488	node	Sl+	0.6	1.2
1	systemd	Ss	0.0	0.3
305	Relay(306)	S	0.0	0.0
51	systemd-journal	S	0.0	0.4
206	unattended-upgr	Ssl	0.0	0.5

Memory Locality

Mean comparison: Row-major 0.017444s vs Column-major 0.217588s — 12.47× ratio



Run	Row-major (s)	Column-major (s)	Ratio
1	0.017528	0.218889	12.49×
2	0.017478	0.223270	12.77×
3	0.017328	0.210606	12.15×
mean	0.017444	0.217588	12.47×

Locality experiment completed successfully.

```
Array size: 4096 x 4096
Total elements: 16777216
Element size: 8 bytes
Array size: 128.00 MiB

run,row_major_s,column_major_s,ratio
1,0.017528,0.218889,12.49
2,0.017478,0.223270,12.77
3,0.017328,0.210606,12.15

Mean row-major time: 0.017444 seconds
Mean column-major time: 0.217588 seconds
Mean ratio: 12.47x

Results saved to results.csv
```